

C4 final practice

1. (Fall 2024, #10) Consider the function F defined on the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by:

$$F(x) = \int_0^x \frac{1}{t^2 + 1} dt$$

You must carefully justify your answers and reasoning.

- (a) Show that $F(x)$ has an inverse on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then determine the value of $(F^{-1})'(0)$.

$$F'(x) = \frac{d}{dx} \left(\int_0^x \frac{1}{t^2 + 1} dt \right) = \frac{1}{x^2 + 1} > 0 \text{ for all } x \text{ in the domain}$$

So F is one-to-one and f^{-1} exists.

$$(F^{-1})'(0) = \frac{1}{F'(F^{-1}(0))} = \frac{1}{F'(0)} = \frac{1}{1} = 1$$

$F^{-1}(0) = 0$ since $F(0) = \int_0^0 \frac{1}{t^2 + 1} dt = 0$

- (b) Let $G(x) = F(\tan(x))$. Calculate the derivative of $G(x)$. Show that $G(x) = x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

$$G'(x) = F'(\tan(x)) \sec^2(x) = \frac{\sec^2(x)}{\tan^2(x) + 1} = \frac{1}{\sin^2(x) + \cos^2(x)} = 1 \text{ for all } x \text{ in the domain}$$

$\Rightarrow G(x) = x + C$ for some constant C (to be determined).

$$\text{Note that } G(0) = F(\tan(0)) = F(0) = 0 \Rightarrow C = 0$$

$$\Rightarrow G(x) = x$$

- (c) Determine the value of $\int_{-1}^1 \frac{1}{t^2 + 1} dt$.

$$\text{From (b), } x = F(\tan(x)) = \int_0^{\tan x} \frac{1}{t^2 + 1} dt \text{ on } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\text{At } x = \frac{\pi}{4}, \tan(x) = 1 \longrightarrow \frac{\pi}{4} = \int_0^1 \frac{1}{t^2 + 1} dt$$

$$x = -\frac{\pi}{4}, \tan(x) = -1 \longrightarrow -\frac{\pi}{4} = \int_0^{-1} \frac{1}{t^2 + 1} dt = -\int_{-1}^0 \frac{1}{t^2 + 1} dt$$

$$\text{So: } \int_{-1}^1 \frac{1}{t^2 + 1} dt = \int_0^1 \frac{1}{t^2 + 1} dt + \int_{-1}^0 \frac{1}{t^2 + 1} dt = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$$

2. (Spring 2024, #9) Consider the following integral function:

$$I(x) = \int_e^{e^x} \frac{1}{t^3 + 3} dt$$

- (a) Find $I'(x)$.

$$\begin{aligned} I'(x) &= \frac{d}{dx} \int_e^{e^x} \frac{1}{t^3 + 3} dt \\ &= \frac{1}{(e^x)^3 + 3} \cdot \frac{d}{dx}(e^x) \\ &= \frac{e^x}{e^{3x} + 3} \end{aligned}$$

- (b) Prove that $I(x)$ is an invertible function.

$$I'(x) > 0 \text{ for all } x \text{ in the domain}$$

$$\Rightarrow I \text{ is one-to-one and invertible}$$

- (c) Find $(I^{-1})'(0)$.

$$\begin{aligned} (I^{-1})'(0) &= \frac{1}{I'(I^{-1}(0))} = \frac{1}{I'(1)} = \frac{1}{\frac{e^1}{e^3 + 3}} = \frac{e^3 + 3}{e} \\ &\quad \uparrow \\ &\quad I^{-1}(0) = 1 \text{ since } I(1) = \int_e^{e^1} \frac{1}{t^3 + 3} dt = 0 \end{aligned}$$

3. (Fall 2023, #9) Given

$$f(x) = 2x + \ln(x^2 - 3)$$

(a) Show that f is one to one on the domain $(\sqrt{3}, +\infty)$.

$$f'(x) = 2 + \underbrace{\frac{2x}{x^2-3}}_{\text{positive for all } x > \sqrt{3}} > 0 \text{ for all } x \text{ in } (\sqrt{3}, +\infty)$$

$\Rightarrow f(x)$ is one-to-one and invertible.

(b) Find $f(2)$.

$$f(2) = 2 \cdot 2 + \ln(2^2 - 3) = 4 + \ln(1) = 4$$

(c) Compute $(f^{-1})'(4)$.

$$\text{From (b), } f^{-1}(4) = 2.$$

$$\text{So } (f^{-1})'(4) = \frac{1}{f'(f^{-1}(4))} = \frac{1}{f'(2)} = \frac{1}{2 + \frac{2 \cdot 2}{2^2 - 3}} = \frac{1}{6}$$

(d) Compute $(f^{-1})'(f(3))$.

$$(f^{-1})'(f(3)) = \frac{1}{f'[f^{-1}(f(3))]} = \frac{1}{f'(3)} = \frac{1}{2 + \frac{2 \cdot 3}{3^2 - 3}} = \frac{1}{3}$$

$\uparrow$
 $f^{-1}(f(x)) = x$ for all x

4. (Spring 2023, #10) The function

$$P(h) = 50 + 13.3 \int_{65}^h e^{-\frac{1}{2}\left(\frac{x-65}{3}\right)^2} dx$$

gives the percent, P , of people in the world that are shorter than h inches tall. For example, a calculator can show that $P(68) = 84$, indicating that 84% of people in the world are shorter than 68 inches tall.

- (a) Show that $P(h)$ has an inverse function. You must use calculus to answer this question, i.e., you cannot just reason from the given context.

$$\begin{aligned} \frac{d}{dh} P(h) &= 13.3 \frac{d}{dh} \int_{65}^h e^{-\frac{1}{2}\left(\frac{x-65}{3}\right)^2} dx \\ &= 13.3 e^{-\frac{1}{2}\left(\frac{h-65}{3}\right)^2} > 0 \quad \text{for all } h \end{aligned}$$

$\Rightarrow P$ is one-to-one and invertible

- (b) If we let $h(P)$ be the inverse of $P(h)$, find $h'(84)$.

$$h'(84) = (P^{-1})'(84) = \frac{1}{P'(P^{-1}(84))}$$

We are given that $P(68) = 84$, so $P^{-1}(84) = 68$

$$\Rightarrow h'(84) = \frac{1}{P'(68)} = \frac{1}{13.3 e^{-\frac{1}{2}\left(\frac{68-65}{3}\right)^2}} = \frac{e^{\frac{1}{2}}}{13.3}$$

- (c) Which of the following are correct interpretations of the quantity $h'(84)$? Circle all that apply.

- A. The slope of the line tangent to $P(h)$ at $(68, 84)$.
- B. The rate of change of P with respect to h at $(68, 84)$.
- ☒ C. The rate of change of h with respect to P at $(84, 68)$.
- D. The approximate share of people in the world whose heights are between 68 and 69 inches.
- ☒ E. The approximate difference in height between someone who is taller than 84% of people in the world and someone who is taller than 85% of people in the world.

5. (Spring 2022, #9) Define $F(x) = \int_1^{x^3} \frac{1}{1+t} dt$ for $x > -1$.

(a) Compute $F'(x)$.

$$F'(x) = \frac{1}{1+x^3} \frac{d}{dx}(x^3) = \frac{3x^2}{1+x^3}$$

(b) Show that F is one-to-one.

For all $x > -1$, we have $x^3 > -1$. So $F'(x) > 0$ for all $x > -1$ (except $F'(0)=0$)
 $\Rightarrow F$ is one-to-one and invertible

(c) Compute $(F^{-1})'(0)$.

$$(F^{-1})'(0) = \frac{1}{F'(F^{-1}(0))} = \frac{1}{F'(1)} = \frac{1}{\frac{3}{1+1}} = \frac{2}{3}$$

$F^{-1}(0)=1$ since $F(1) = \int_1^1 \frac{1}{1+t} dt = 0$